

Flyback SMPS Design Session — Detailed Engineering Notes

Comprehensive design rationale, component selection tables, and bring-up guidance

1. System Overview

Topology: Isolated flyback converter

Controller: Power Integrations TNY285 (TinySwitch family)

Nominal Output: 5 V @ 1 A

Feedback: TL431 + optocoupler (secondary-side regulation)

Bias Supply: Auxiliary transformer winding feeding BP pin

Primary goals:

- Match known-good deadbug prototype
- Ensure prefab / SMT manufacturability
- Prioritize stability, EMI robustness, and margin

2. Primary-Side Clamp Network (RCD + Z)

Purpose:

- Limit drain overvoltage from leakage inductance
- Protect internal MOSFET
- Shape dissipation and EMI

Architecture:

- RCD clamp dissipates bulk energy
- TVS diode provides hard voltage ceiling

Key insight:

Small-signal zeners are NOT appropriate for drain clamping.

A true TVS diode is required due to pulse energy.

Element	Chosen Class	Reasoning
Clamp Diode	Ultrafast PN Rectifier (UF class)	High VRRM, fast recovery
Clamp Resistor	1206 / axial metal film	Dissipation shaping
Clamp Capacitor	C0G / film (nF range)	Low loss
Z Clamp	TVS (SMBJ250A class)	Handles high-energy spikes

3. Output Rectification and Filtering

Output Rectifier:

- Schottky diode preferred
- Low forward drop
- No reverse recovery

Rating rule:

- Voltage: 40–60 V
- Current: ≥ 3 A (pulsed flyback current)

Filtering philosophy:

- Bulk cap provides low-frequency energy
- Ceramics handle high-frequency noise
- ESR contributes damping

Component	Value / Class	Notes
Bulk Capacitor	470 μ F hybrid polymer, 16–35 V	Low ESR, high ripple
HF Ceramic	1 μ F X7R	Placed near diode
HF Ceramic	0.1 μ F X7R	EMI suppression

4. TL431 + Optocoupler Feedback Loop

TL431 acts as the error amplifier.

Optocoupler transfers error signal across isolation.

TNY285 adjusts delivered energy per cycle.

Critical point:

Optocoupler regulates OUTPUT voltage, not BP pin.

Compensation Network

Compensation capacitor placement:

- MUST be between TL431 REF and CATHODE

Not allowed:

- Across optocoupler LED
- From REF to ANODE

Typical value:

- 1 nF (C0G)

Optional tuning range:

- 1–4.7 nF

5. Auxiliary Winding and BP Pin Supply

Function:

- Auxiliary winding powers the controller after startup
- Reduces internal dissipation
- Improves efficiency and stability

This is NOT a regulation loop.

It is a power supply tap.

Component	Typical Value	Purpose
Aux Diode	Fast rectifier or Schottky	Rectify aux voltage
BP Capacitor	10 μ F, ≥ 25 V	Energy reservoir
Bleeder Resistor	15 k Ω	Predictable discharge

6. Input / DC-Bus π -Filter Inductor

Role:

- Attenuate rectified bus ripple
- Reduce conducted EMI

Correct class:

- mH-range inductance
- Low current rating
- High DCR acceptable

Incorrect class:

- μ H buck/output inductors

Parameter	Target
Inductance	≈ 1 mH
Current Rating	$\geq 0.3\text{--}0.5$ A
Package	SMD drum / EMI choke

7. Manufacturability Rules Applied

- SMT parts preferred
- Tape-and-reel only
- Avoid NRND families
- Common packages (SOT-23, SMB, 1206, SMD cans)
- DNI footprints added for tuning

8. Bring-Up Expectations

Startup:

- BP initially charged from internal HV source
- Switching begins
- Aux winding takes over BP

Normal operation:

- Output regulated at 5 V by opto loop
- BP ~12–18 V from aux winding
- Stable, cool operation